PREP 2 · LESSONS 3–4

Potential difference and resistance

Voltmeters · energy per coulomb, E = Q V · resistance, V = I R

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define electric current. DN 90 [1]

The rate of flow of electrical charge.

2 State the standard units of charge, current and time. DN 91 [1]

Charge: coulomb (C); current: ampere (A); time: second (s)

3 State the function of a fuse. DN 81 [1]

To melt and break the circuit if the current becomes too large.

Key ideas LESSONS 3–4

Show your working

4 A voltmeter is connected in **series** (**parallel**) with the component it measures. [1]

5 15 C of charge passes through a lamp. The potential difference across the lamp is 12 V. Use E = Q V

Calculate the energy transferred. [2]

E = Q V = 15 × 12 ✓

E = 180 J ✓

6 The potential difference across a resistor is 6.0 V. The current through it is 0.30 A. Use V = I R

Calculate the resistance of the resistor. [3]

V = I R, so R = V ÷ I ✓

R = 6.0 ÷ 0.30 ✓

R = 20 Ω ✓

7 For an ohmic conductor at a constant temperature, the current is [1]

(**directly proportional**) | **inversely proportional** | **not related** to the potential difference.